Calculus: Differentiation and Continuity

Challenging Practice — 45 Minutes

Name: _____ Date: _____

Instructions

Complete all 12 problems without a calculator. Allow about 22 minutes for Problems 1–6 and 23 minutes for Problems 7–12. Show derivative steps, limit calculations, and theorem hypotheses where requested. Give exact answers; factored derivatives are acceptable. All angles are in radians. Do not use L'Hôpital's rule or Taylor series. This cover sheet may be used throughout the practice.

Basic Formulas and Facts

Derivative definition and rules

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

$$(u+v)' = u' + v', \qquad (cu)' = cu',$$
$$(u^n)' = nu^{n-1}u',$$
$$(uv)' = u'v + uv',$$
$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \quad v \neq 0,$$
$$[f(g(x))]' = f'(g(x))g'(x).$$

Here u and v are differentiable functions of x, and c is constant.

Common derivatives

$$(e^u)' = e^u u', \qquad (\ln u)' = \frac{u'}{u} \ (u > 0),$$
$$(\sin u)' = \cos(u)u',$$
$$(\cos u)' = -\sin(u)u',$$
$$(\tan u)' = \sec^2(u)u'.$$

Implicit differentiation

$$\frac{d}{dx}(y^n) = ny^{n-1}y', \qquad \frac{d}{dx}(xy) = y + xy'.$$

For $F(x, y) = C$, if $F_y \neq 0$,

$$y' = -\frac{F_x}{F_y}.$$

Here F_x and F_y mean differentiation with respect to one variable while holding the other fixed. A horizontal tangent requires $F_x = 0$, $F_y \neq 0$; a vertical tangent requires $F_y = 0$, $F_x \neq 0$.

Tangent line and second derivative

$$y - f(a) = f'(a)(x - a), \qquad y'' = \frac{d}{dx}(y').$$

Limits

Use limit laws when the required limits exist; a quotient law requires a nonzero denominator limit. For continuous functions, substitute directly.

$$\lim_{t\to 0} \frac{\sin t}{t} = 1, \qquad \lim_{t\to 0} \frac{\tan t}{t} = 1,$$
$$\lim_{t\to 0} \frac{1 - \cos t}{t^2} = \frac{1}{2}.$$

$$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

Continuity and differentiability

Continuity at a requires $f(a)$ to be defined and

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a).$$

For a piecewise join, differentiability requires continuity and equal, finite one-sided difference-quotient limits. Differentiability implies continuity; continuity does not imply differentiability. Existence of $f'(a)$ does not require f' to be continuous at a.

Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and N lies strictly between $f(a)$ and $f(b)$, then $f(c) = N$ for some $c \in (a, b)$. This does not guarantee uniqueness.

Mean Value Theorem (MVT)

If f is continuous on $[a, b]$ and differentiable on (a, b), then some $c \in (a, b)$ satisfies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Rolle's theorem

Under the MVT hypotheses, if $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

A failed hypothesis means the theorem cannot be invoked; its conclusion might still happen to hold.

Part I: Chain Rule and Implicit Differentiation

Problems 1–6: approximately 22 minutes total

1. Differentiate.

$$y = \frac{e^{\sin(3x)} \ln(1 + x^2)}{\sqrt{2 + x^4}}$$

2. Differentiable functions f and g satisfy

$$g(0) = 2, \qquad g'(0) = -1, \qquad f(5) = 3, \qquad f'(5) = 4.$$

For

$$H(x) = \ln\Big(1 + \big[f\big(g(x)^2 + 1\big)\big]^2\Big),$$

find $H'(0)$ and the tangent line at $x = 0$.

3. For $y = \sin(x^2)$, find y'' and evaluate it at $x = \sqrt{\pi/2}$.

Part I — Continued

4. For the curve
$$e^{xy} + x^2 - y^2 = 2,$$
find dy/dx in terms of x and y, and find the tangent line at $(1, 0)$.

5. For the curve
$$x^2y + y^3 = 2,$$
find dy/dx in terms of x and y, and find d^2y/dx^2 at $(1, 1)$.

6. Find all points on
$$x^2 + xy + y^2 = 12$$
where the tangent is horizontal or vertical. Justify each classification and check that the points lie on the curve.

Part II: Limits, Continuity, and Theorems

Problems 7–12: approximately 23 minutes total

7. Evaluate both limits, showing the transformations used.

$$\text{(a)} \quad \lim_{x \to 0} \frac{\sqrt{1+4x}-1}{\sin(3x)} \qquad \text{(b)} \quad \lim_{x \to 0} \frac{1-\cos(4x)}{x\sin(2x)}$$

8. Determine all real values of a, b, and k for which p is differentiable at $x = 2$. Then find $p'(2)$. Show the continuity and differentiability conditions separately.

$$p(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x < 2, \\ k, & x = 2, \\ ax^2 + b, & x > 2. \end{cases}$$

9. Let

$$q(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Determine whether q is continuous at 0, whether $q'(0)$ exists, and whether q' is continuous at 0. Use the derivative definition to establish $q'(0)$, and justify any nonexistent limit.

Part II — Continued

10. Does Rolle's theorem apply to $r(x) = |x^2 - 1|$ on $[-2, 2]$? Check every hypothesis. Regardless of your answer, find all $c \in (-2, 2)$ for which $r'(c) = 0$.

11. Let

$$s(x) = \begin{cases} x^2, & x \le 1, \\ 2x - 1, & x > 1. \end{cases}$$

Verify every hypothesis of the Mean Value Theorem on $[0, 3]$ and find all $c \in (0, 3)$ satisfying its conclusion.

12. For real a, define

$$t_a(x) = \begin{cases} x^2 + 2, & x \le 0, \\ ax + 2, & x > 0. \end{cases}$$

Determine all values of a for which the Intermediate Value Theorem, applied to the endpoints of $[-1, 2]$, guarantees a zero in $(-1, 2)$. For those values, find the zero and determine whether it is unique. Justify the theorem's hypotheses and your uniqueness claim.